AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY

---AMITY UNIVERSITY JHARKHAND, RANCHI---

Student Name: Palak Goenka **Program:** B.Tech CSE (AI & ML), Sem–V

Enroll No.: A357175124008

Title: Early Detection of Academic Burnout using Machine Learning

NTCC Weekly Progress Report (Minor Project): Wk. No: 1 *Duration: 06/08/2026 – 14/08/2026*

DAYS / Date	Summary
THU / 06/08/2026	Reviewed NTCC project guidelines, finalized problem statement, and outlined project deliverables for academic burnout detection.
FRI / 07/08/2026	Studied psychological burnout frameworks, including the Maslach Burnout Inventory (MBI-SS) and Job Demands-Resources (JD-R) model.
MON / 10/08/2026	Researched machine learning applications in educational data mining, cognitive fatigue, and student well-being monitoring.
TUE / 11/08/2026	Explored behavioral feature representation across academic workload, sleep patterns, screen time, and psychosocial indicators.
WED / 12/08/2026	Listed and evaluated required tools, libraries (Python, Scikit-Learn, Pandas, NumPy, Seaborn), and environment dependencies.
THU / 13/08/2026	Drafted initial system architecture blueprint and prepared roadmap for data preprocessing, modeling, and evaluation.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 2 *Duration: 15/08/2026 – 23/08/2026*

DAYS / Date	Summary
MON / 17/08/2026	Acquired multi-institutional survey dataset comprising 3,000 verified university student records across 18 behavioral attributes.
TUE / 18/08/2026	Conducted data cleaning, verified zero null/missing values, and inspected variable data types using Pandas.
WED / 19/08/2026	Analyzed target variable distribution across Low (41.4%), Medium (47.7%), and High (10.9%) burnout tiers.
THU / 20/08/2026	Configured categorical encoding using OneHotEncoder with drop-first settings for gender, tuition, and university type.
FRI / 21/08/2026	Configured numerical feature standardization using StandardScaler within an integrated ColumnTransformer pipeline.
SAT / 22/08/2026	Created reproducible preprocessing scripts and partitioned dataset into 80% training and 20% held-out test splits.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 3 *Duration: 24/08/2026 – 01/09/2026*

DAYS / Date	Summary
MON / 24/08/2026	Computed Pearson correlation matrix to quantify statistical dependencies between lifestyle predictors and stress score.
TUE / 25/08/2026	Generated correlation matrix heatmap plot to visualize collinearity among academic and mental health variables.
WED / 26/08/2026	Plotted stratified boxplots analyzing the distribution of anxiety, sleep hours, and screen time across burnout tiers.
THU / 27/08/2026	Formulated and coded the Academic Pressure Index (API = Assignment Load × Exam Frequency) interaction feature.
FRI / 28/08/2026	Formulated and coded Screen-to-Sleep Ratio (SSR) and Social Support Deficit (SSD = Peer Pressure - Family Support).
MON / 31/08/2026	Implemented Lifestyle Balance Score (LBS) combining sleep hours, physical exercise, and screen time deductions.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 4 *Duration: 02/09/2026 – 10/09/2026*

DAYS / Date	Summary
WED / 02/09/2026	Configured 5-Fold Stratified Cross-Validation schema to prevent data leakage and handle class imbalance.
THU / 03/09/2026	Implemented and evaluated baseline distance-based K-Nearest Neighbors (KNN) classification algorithm.
SAT / 05/09/2026	Developed and trained Extremely Randomized Trees (Extra Trees) classifier with 150 decision trees.
MON / 07/09/2026	Constructed Random Forest ensemble classifier with 150 estimators and constrained max depth of 12.
TUE / 08/09/2026	Evaluated ensemble bagging performance and analyzed Gini impurity splits across decision trees.
WED / 09/09/2026	Implemented Gradient Boosting Classifier (GBM) with 120 estimators and sequential pseudo-residual learning.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 5 *Duration: 11/09/2026 – 19/09/2026*

DAYS / Date	Summary
FRI / 11/09/2026	Implemented Support Vector Machine (SVM) utilizing non-linear Radial Basis Function (RBF) kernel.
SAT / 12/09/2026	Configured RBF kernel hyperparameter tuning across cost parameter C and gamma scaling factors.
MON / 14/09/2026	Constructed Multi-Layer Perceptron (MLP) neural network with two hidden layers (64, 32 units) and ReLU activations.
TUE / 15/09/2026	Configured Adam optimizer and backpropagation routines for MLP neural network training.
WED / 16/09/2026	Implemented Multinomial Logistic Regression model with L2 regularization and softmax link function.
THU / 17/09/2026	Executed 5-fold cross-validation benchmarking across all seven candidate classification architectures.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 6 *Duration: 20/09/2026 – 28/09/2026*

DAYS / Date	Summary
MON / 21/09/2026	Benchmarked cross-validation results and identified Multinomial Logistic Regression as optimal model with 99.50% accuracy.
TUE / 22/09/2026	Generated comparative performance bar chart visualizing Accuracy and Macro F1-score across all 7 models.
WED / 23/09/2026	Evaluated the optimal classifier on the 600-student test split and computed Precision, Recall, and Macro F1 (0.9909).
THU / 24/09/2026	Generated absolute count and normalized percentage confusion matrices for the test evaluation split.
FRI / 25/09/2026	Developed continuous regression models for predicting Stress_Score; trained Ridge Regression, Random Forest, and GBM regressors.
SAT / 26/09/2026	Validated Ridge Regression achieving $R^2 = 1.0000$ (MAE: 0.00, RMSE: 0.01) and plotted regression parity scatter plot.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 7 *Duration: 29/09/2026 – 07/10/2026*

DAYS / Date	Summary
TUE / 29/09/2026	Extracted model coefficient weights and ranked top 14 predictive features driving student academic burnout.
WED / 30/09/2026	Plotted feature importance bar chart identifying assignment load, exam frequency, and sleep hours as primary drivers.
THU / 01/10/2026	Designed rule-based prescriptive intervention logic to deliver personalized lifestyle and academic counseling recommendations.
SAT / 03/10/2026	Built standalone CLI inference script (predict.py) and serialized trained models into persistent joblib binary artifacts.
MON / 05/10/2026	Developed Flask web application backend (app.py) supporting REST API endpoints for single and batch predictions.
TUE / 06/10/2026	Configured batch CSV upload processor with dynamic feature engineering and bulk prediction output handling.

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NTCC Weekly Progress Report (Minor Project): Wk. No: 8 *Duration: 08/10/2026 – 15/10/2026*

DAYS / Date	Summary
THU / 08/10/2026	Designed responsive glassmorphic web dashboard UI (index.html) with interactive sliders, badges, and charts.
FRI / 09/10/2026	Wrote comprehensive unit and integration test suite (test_model.py) verifying data integrity, feature math, and APIs.
SAT / 10/10/2026	Executed automated test suite, confirming all 7 unit tests passed successfully in 0.145 seconds.
MON / 12/10/2026	Compiled 17-page formal academic thesis document matching Amity University Jharkhand template with embedded figures.
WED / 14/10/2026	Deployed web application to Vercel cloud platform (academic-burnout.vercel.app) and synced GitHub repository.
THU / 15/10/2026	Finalized NTCC weekly progress reports, validated all project deliverables, and completed report submission.

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